

**This Is Not the ex-Gaussian Model You Are Looking For: On the Default Parameterization of
Bayesian ex-Gaussian Models in brms**

Jason Geller

Department of Psychology and Neuroscience, Boston College

Author Note

Jason Geller  <https://orcid.org/0000-0002-7459-4505>

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Correspondence concerning this article should be addressed to Jason Geller, Department of Psychology and Neuroscience, Boston College, Email: drjasongeller@gmail.com

Abstract

The ex-Gaussian distribution is a popular mathematical model for analyzing reaction-time (RT) data because it separates the central portion of the distribution, captured by the Gaussian parameters μ and σ , from the positively skewed tail, captured by the exponential parameter τ . Researchers often use these parameters to make inferences about underlying cognitive processes. As Bayesian methods have become increasingly accessible, complex ex-Gaussian models can now be fit with relative ease. In R, the {brms} package makes this especially straightforward. However, it is important to recognize that the default parameterization offered by {brms} does not map onto the classical parameterization familiar to many researchers in the cognitive sciences. As a result, some published studies may have drawn incorrect inferences from effects estimated on the default μ parameter. In the default {brms} model, μ indexes the mean of the full ex-Gaussian distribution rather than the location of the Gaussian component alone. This distinction matters because changes in the Gaussian location and changes in the exponential tail can offset one another at the level of the overall mean. Using simulated data in which blurred and clear words have Gaussian locations and exponential tail shifts that move in opposite directions—holding the overall distributional mean constant—I show that the default parameterization returns a null effect on μ , whereas the classical parameterization recovers the true shift in the Gaussian component. I then demonstrate how to fit the classical parameterization directly in {brms}. My hope is that this commentary will help researchers and reviewers better understand the models they are fitting, ensure that model parameters match their theoretical questions, more accurately evaluate ex-Gaussian models reported in the literature.

Keywords: ex-Gaussian, reaction time, Bayesian estimation, brms, parameterization

This Is Not the ex-Gaussian Model You Are Looking For: On the Default Parameterization of Bayesian ex-Gaussian Models in brms

Over the past two decades, psychology has seen a substantial increase in the application of Bayesian data analysis methods as a supplement to widely used frequentist methods (Pfadt et al., 2025). This shift has been driven by the increasing availability of computational tools that have made fitting Bayesian models both practical and efficient. Programming languages like R (R Core Team, 2024), Python (Van Rossum & Drake, 2009), and Julia (Bezanson et al., 2017) provide libraries and packages that allow researchers to fit Bayesian models with little technical overhead. In addition, growing dissatisfaction with null-hypothesis significance testing (NHST) and its associated pitfalls (e.g., p-value misinterpretation and dichotomous thinking), together with the replication crisis/credibility revolution (Vazire, 2018), has pushed many researchers toward statistical alternatives.

One domain that has seen an increase in the adoption of Bayesian methods is the analysis of response time (RT) data in cognitive sciences where RTs are routinely used as measures of processing speed, decision difficulty, and cognitive efficiency. Traditionally, RT data have often been analyzed using summary statistics such as the mean or median. However, RT distributions are typically positively skewed, bounded at zero, and characterized by substantial variability both within and between individuals (Wilcox & Keselman, 2003). Moreover, experimental manipulations may affect not only the central tendency of the distribution but also its variability or upper tail. Consequently, analyses based solely on the mean or median can obscure theoretically meaningful distributional effects, motivating calls to move beyond mean-based analyses of RT data (Balota & Yap, 2011).

One way to move beyond the mean is to model the full shape of the RT distribution. Mathematical models provide a flexible way to do so, allowing researchers to draw inferences about the psychological processes—such as memory, attention, and language—that give rise to observed response times. One model that has become especially popular for this purpose is the ex-Gaussian model (Ratcliff & Murdock, 1976). The ex-Gaussian distribution is formed by the convolution of a Gaussian distribution and an exponential distribution and is characterized by three parameters: μ , σ , and τ . The parameter μ represents the mean of the Gaussian component, whereas σ represents its standard deviation. The τ parameter represents both the mean and standard deviation of the exponential component and primarily governs the length of the distribution's positively skewed upper tail. The overall mean of the ex-Gaussian distribution is $\mu + \tau$, and the standard deviation (SD) is $\sqrt{\sigma^2 + \tau^2}$. Although these parameters have been used to infer cognitive processes such as automaticity (Balota et al., 2008; Dewitt

et al., 2018) and cognitive control/inhibition (Kane & Engle, 2003), these interpretations are not inherent to the parameters and should be made cautiously (Matzke & Wagenmakers, 2009). Nevertheless, the ex-Gaussian model can provide a more complete description of RT data than the arithmetic mean alone, which can be disproportionately influenced by the positively skewed tail of the distribution.

The ex-Gaussian model can be fit within a Bayesian framework relatively easily using the {brms} package in R which provides a high-level interface to the probabilistic programming language Stan (Carpenter et al., 2017). The {brms} package supports a wide range of response distributions, including the ex-Gaussian family, and allows researchers to incorporate complex model structures at both the participant and item levels as well as across different parameters. This accessibility has contributed to the growing use of Bayesian ex-Gaussian models in reaction-time research (see Angele et al., 2023, 2024; Geller et al., 2025; Kinoshita & Liong, 2023; Serrano-Carot et al., 2026). However, an important complication arises from the default parameterization used by the R package {brms}. Although the model is straightforward to estimate, its default specification does not necessarily provide direct access to the parameter-specific comparisons researchers often wish to make.

The central problem is that the parameter labeled μ in the default {brms} ex-Gaussian model does not correspond to the mean of the Gaussian component in the conventional parameterization. Instead, the quantity modeled as μ in {brms} represents the mean of the full ex-Gaussian distribution. Under the conventional parameterization, this quantity is equal to $\mu_{\text{Gaussian}} + \tau$. Consequently, regression effects estimated for the default μ parameter reflect changes in a composite quantity that combines the Gaussian location and exponential tail components. Such effects therefore cannot be interpreted directly as changes in the Gaussian component or in typical response speed. To recover the conventional Gaussian mean, it is necessary to derive it from the model parameters as $\mu_{\text{Gaussian}} = \mu_{\text{brms}} - \tau$ or to specify an alternative parameterization that models the conventional ex-Gaussian parameters directly.

In this commentary, we demonstrate what the default parameterization of the ex-Gaussian model in {brms} actually provides and then show how to specify the model so that its parameters correspond to the conventional ex-Gaussian parameters used in reaction-time research. The goal is to improve the accuracy and interpretability of inferences drawn from these models. This issue is particularly important because the ex-Gaussian distribution is widely used to distinguish changes in the central portion of the RT distribution from changes in its positively skewed tail.

Simulation Example

To demonstrate the issue with fitting the default ex-Gaussian model in {brms}, I simulate data using a comparison adapted from one of my own studies (Geller et al., 2025) in R [version 4.6.0; R Core Team (2024)]. In Geller et al. (2025), we examined how perceptual disfluency—manipulated by varying the level of blur applied to letter strings influenced reaction times in a lexical decision task (i.e., is this letter string a word?), and whether these effects were related to how well someone remembers the words in a recognition memory task. In that study, low blur words affected early stages of encoding, reflected in the Gaussian location parameter μ , but not the later, tail component captured by τ . For the present illustration, however, I construct a sharper scenario in which μ and τ differ in opposite directions so that the two conditions share the same expected reaction time. Using the `rexgaussian2()` function from the {brms.exgaussian} package, I simulate ex-Gaussian data for clearly presented and low-blur words in which μ is set to 500 ms and 550 ms, τ to 100 ms and 50 ms, and σ to 60 ms in both conditions (see Table 1).

Table 1

Data-generating ex-Gaussian parameters for the Clear and Low-Blur conditions, with opposing changes in μ and τ that hold the expected response time ($\mu + \tau$) constant.

Blur	μ	σ	τ	Expected RT
Clear	500	60	100	600
Low-Blur	550	60	50	600

Simulate the Data

We first draw 100 reaction times per condition (200 in total) from the ex-Gaussian distributions defined by these parameters.

```

1  # Fix the random seed so the simulated data are identical on every render.
2  set.seed(1234)
3
4  # Number of simulated trials per condition (200 observations in total).
5  n_per_group <- 100
6
7  # One row per trial, labelled by Blur condition (Clear is the reference level).
8  d <- data.frame(

```

```

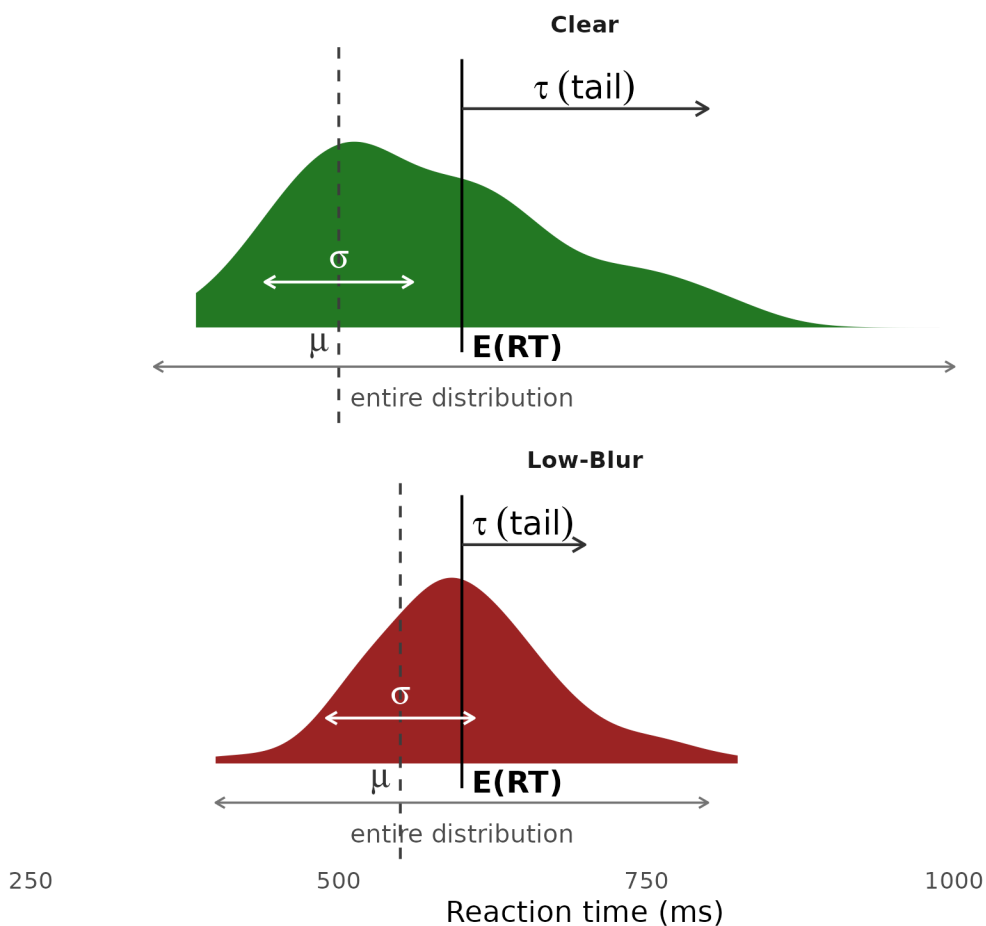
128 9     Blur = factor(
129 10       rep(c("Clear", "Low-Blur"), each = n_per_group),
130 11       levels = c("Clear", "Low-Blur")
131 12     )
132 13 )
133 14
134 15 # Row index (1 = Clear, 2 = Low-Blur) used to look up each trial's parameters.
135 16 parameter_index <- as.integer(d$Blur)
136 17
137 18 # Draw reaction times from the ex-Gaussian using each trial's condition values.
138 19 d$rt <- brms.exgaussian::rexgaussian2(
139 20   n = nrow(d),
140 21   mu = sim_params$mu[parameter_index],
141 22   sigma = sim_params$sigma[parameter_index],
142 23   tau = sim_params$tau[parameter_index]
143 24 )

```

Figure 1 shows the simulated RT distribution for each condition and how the three ex-Gaussian parameters shape it: μ locates the Gaussian component, σ sets its spread, and τ stretches the right tail. Because the increase in μ from Clear to Low-Blur is offset by an equal decrease in τ , the overall mean $E(RT) = \mu + \tau$ is identical (600 ms) in both conditions.

Figure 1

Simulated reaction-time distributions for each condition, annotated with the ex-Gaussian parameters. The dashed line marks the Gaussian location μ ; the double arrow spans $\pm\sigma$ around it (the Gaussian spread); the rightward arrow over the tail marks the exponential component τ , which stretches the distribution to the right; and the solid line marks the overall mean $E(RT) = \mu + \tau$, which is 600 ms in both conditions. The bottom span highlights that $E(RT)$ is the mean of the entire distribution — the quantity the default *brms* μ estimates.

**Default {brms} Parameterization**

Using this data we fit the default *brms* ex-Gaussian model. In this parameterization, the parameter labeled μ represents the expected value of the full ex-Gaussian distribution rather than the location of its Gaussian component.

The exponential component is labeled *beta*. On the response scale, *beta* corresponds to the classical ex-Gaussian parameter τ :

$$\beta = \tau_{\text{classical}}.$$

Thus, `beta` and τ are not substantively different quantities. Both represent the mean of the exponential component. The difference is primarily one of notation. The important difference between the default {} model and the classical parameterization concerns `mu`:

$$\mu = \mu_{\text{classical}} + \tau.$$

In other words, the default {} parameterization uses `mu` for the expected value of the complete ex-Gaussian distribution, whereas the classical parameterization uses μ for the location of the Gaussian component.

Both the expected response and the exponential component are allowed to vary across conditions.

Prior Specification

The priors were selected to reflect the known scale of the simulated data while remaining sufficiently broad to allow the model to recover deviations from the generating values.

```

1  priors_default <- c(
2    # mu intercept: centered at 600 ms, the expected RT (E(RT)) of the reference
3    # condition in this default parameterization.
4    prior(
5      normal(600, 100),
6      class = "Intercept"
7    ),
8    # Effect of Blur on mu: centered at 0 (no assumed direction).
9    prior(
10     normal(0, 50),
11     class = "b"
12   ),
13   # beta (exponential mean) intercept, on the log scale: exp(log(100)) = 100 ms.
14   prior(
15     normal(log(100), 0.5),
16     class = "Intercept",
17     dpar = "beta"

```

```

190 18  ),
191 19  # Effect of Blur on beta (log scale): centered at 0 (a ratio of 1).
192 20  prior(
193 21    normal(0, 0.5),
194 22    class = "b",
195 23    dpar = "beta"
196 24  )
197 25  )

```

198 The intercept prior for the primary `mu` submodel is

$$\mu_{\text{Intercept}} \sim \mathcal{N}(600, 100).$$

199 We center this prior at 600 ms because, in the default parameterization, `mu` represents the expected RT.

200 The simulated expected RT for the reference condition is

$$500 + 100 = 600 \text{ ms}.$$

201 The standard deviation of 100 ms makes the prior informative about the plausible scale of RTs without
 202 tightly constraining the intercept to equal exactly 600 ms.

203 The prior on the `Blur` coefficient is

$$b_{\text{Blur}} \sim \mathcal{N}(0, 50).$$

204 This prior is centered at zero because, before observing the data, positive and negative condition effects
 205 are both plausible. A standard deviation of 50 ms allows substantial differences between conditions
 206 while placing less prior mass on implausibly large effects.

207 The `beta` parameter is positive and uses a log link. Its intercept is therefore modeled on the log
 208 scale:

$$\log(\beta_{\text{Clear}}) \sim \mathcal{N}(\log(100), 0.5).$$

209 The prior is centered at $\log(100)$ because the simulated value of the exponential mean for the reference
 210 condition is 100 ms. After applying the inverse-link transformation, the prior is centered approximately
 211 around

$$\beta_{\text{Clear}} = 100 \text{ ms}.$$

212 The standard deviation of 0.5 is on the log scale. It therefore represents multiplicative rather than
 213 additive uncertainty.

214 The prior on the effect of `Blur` on `beta` is

$$b_{\beta, \text{Blur}} \sim \mathcal{N}(0, 0.5).$$

215 Because this coefficient is also on the log scale, a value of zero represents no difference between
216 conditions:

$$\exp(0) = 1.$$

217 The simulated change from 100 ms to 50 ms corresponds to the log ratio

$$\log\left(\frac{50}{100}\right) = \log(0.5) \approx -0.693.$$

218 The prior $\mathcal{N}(0, 0.5)$ places reasonable prior mass near this value while still favoring smaller differences.

219 The model can then be fit as follows:

```
220 1  fit_brms <- brm(
221 2    bf(
222 3      rt ~ Blur,      # mu submodel: here mu is E(RT), not the Gaussian location
223 4      sigma ~ 1,      # constant Gaussian SD
224 5      beta ~ Blur     # exponential mean (the classical tau), allowed to vary
225 6    ),
226 7    data = d,
227 8    prior = priors_default,
228 9    family = exgaussian(), # 's built-in ex-Gaussian family
229 10   chains = 2,
230 11   iter = 2000,
231 12   warmup = 500,
232 13   backend = "cmdstanr",
233 14   cores = 4,
234 15   seed = 1234          # reproducible draws
235 16 )
```

236 *Estimating Ex-Gaussian Parameters with {easystats}*

237 The `{modelbased}` package, which is part of the `{easystats}` ecosystem, can easily be used to
238 obtain condition-specific posterior estimates from a fitted model. The `estimate_means()` function

returns estimated parameters for each condition, back-transformed, so estimates are presented on time scale, whereas `estimate_contrasts()` computes differences between conditions.

As shown in Table 1, the present data were generated so that the two conditions have the same expected reaction time but differ in the Gaussian and exponential components of the ex-Gaussian distribution.

Under the classical ex-Gaussian parameterization, the expected response is

$$E(RT) = \mu + \tau.$$

Thus, the 50-ms increase in μ for the Low-Blur condition is offset by a 50-ms decrease in τ . The expected RT is therefore 600 ms in both conditions.

Estimating the Expected Response (μ)

Condition-specific estimates of the expected response can be obtained with `estimate_means()` and setting `predict` argument to “mu”:

```
1 # Default model: per-condition mu, which equals E(RT) in this parameterization.
2 overall_means <- estimate_means(
3   fit_brms,
4   by = "Blur",
5   predict = "mu",
6   centrality = "mean"
7 )
```

Table 2

Estimated expected response, $E(RT)$, by condition from the default model.

Blur	Mu	95% CrI	pd
Clear	580.2	[558.3, 604.9]	1
Low-Blur	603.0	[587.9, 619.3]	1

Because the primary `mu` parameter in the default ex-Gaussian model represents $E(RT)$, these estimates should be close to 600 ms in both conditions.

The corresponding contrast can be obtained using `estimate_contrasts()`:

```
1 # Default model: Blur contrast in E(RT) ( $\approx 0$  by design).
2 overall_difference <- estimate_contrasts(
```

```

267 3   fit_brms,
268 4   contrast = "Blur",
269 5   centrality = "mean"
270 6   )

```

271 Table 3

272 *Blur contrast in the expected response from the default brms model.*

273	Contrast	Mean	95% CrI	pd
274	Low-Blur - Clear	22.8	[-5.3, 49.1]	0.949

275 The posterior estimate of this contrast should be close to zero because the two conditions were
 276 generated with the same expected RT:

$$(550 + 50) - (500 + 100) = 600 - 600 = 0.$$

277 Thus, the default model correctly recovers the absence of a difference in the full distributional mean.

278 However, its primary `mu` coefficient does not directly reveal the 50-ms difference in the Gaussian-
 279 location parameter.

280 **Estimating the Exponential Component.** In the default `{brms}` parameterization, `beta`
 281 represents the mean of the exponential component. It therefore corresponds directly to τ in the classical
 282 parameterization:

$$\beta = \tau.$$

283 Condition-specific estimates can be obtained by requesting predictions for `beta`:

```

284 1   # Default model: per-condition beta (the exponential mean = classical tau).
285 2   tau_estimates_default <- estimate_means(
286 3     fit_brms,
287 4     by = "Blur",
288 5     predict = "beta",
289 6     centrality = "mean"
290 7   )

```

Table 4

*Estimated exponential component (*beta*) by condition from the default brms model.*

Blur	Beta	95% CrI	pd
Clear	107.2	[79.8, 140.8]	1
Low-Blur	55.9	[35.5, 79.4]	1

Although the resulting object is called `tau_estimates_default` for interpretive clarity, the parameter is named `beta` inside the default `brm` model.

These estimates should be approximately 100 ms for Clear and 50 ms for Low-Blur.

The corresponding contrast can be obtained using:

```
1 # Default model: Blur contrast in beta (the exponential component).
2 tau_difference_default <- estimate_contrasts(
3   fit_brms,
4   contrast = "Blur",
5   predict = "beta",
6   centrality = "mean"
7 )
```

Table 5

*Blur contrast in the exponential component (*beta*) from the default brms model.*

Contrast	Mean	95% CrI	pd
Low-Blur - Clear	-51.3	[-86.5, -20.6]	0.999

As we can see in [Table 5](#), the difference between the two conditions is approximately 50 ms.

Classical Ex-Gaussian Parameterization

As we have shown above, the default parameterization gives you the expected RT of the entire distribution. We can instead fit the model using the `{brms.exgaussian}` package, which implements the classical ex-Gaussian parameterization. In this model, μ represents the location of the Gaussian component and τ represents the mean of the exponential component.

We first define the custom family and Stan functions from the `{brms.exgaussian}` package:

```
1 # Custom family that reparameterizes brms so mu is the Gaussian location.
2 exg_stanvars <- exgaussian2_stancode() # Stan code injected into the model
3 exg_family <- exgaussian2()           # the custom family definition
```

Prior Specification

The priors for the classical model are specified as follows:

```
1  priors_classical <- c(
2    # mu intercept: centered at 500 ms, now the Gaussian location of the
3    # reference condition (not E(RT)).
4    prior(
5      normal(500, 100),
6      class = "Intercept"
7    ),
8    # Effect of Blur on mu (Gaussian location): centered at 0.
9    prior(
10     normal(0, 50),
11     class = "b"
12   ),
13   # tau (exponential mean) intercept, log scale: exp(log(100)) = 100 ms.
14   prior(
15     normal(log(100), 0.5),
16     class = "Intercept",
17     dpar = "tau"
18   ),
19   # Effect of Blur on tau (log scale): centered at 0.
20   prior(
21     normal(0, 0.5),
22     class = "b",
23     dpar = "tau"
24   ),
25   # sigma (Gaussian SD) intercept, log scale: exp(log(60)) = 60 ms.
26   prior(
27     normal(log(60), 0.5),
28     class = "Intercept",
```

```

351 29      dpar = "sigma"
352 30    )
353 31  )

```

These priors follow the same reasoning as those for the default model (described above), with two changes. First, the intercept is now centered at 500 ms rather than 600 ms, because in the classical parameterization `mu` denotes the Gaussian-location parameter (500 ms for the Clear reference condition) rather than the expected RT:

$$\mu_{\text{Intercept}} \sim \mathcal{N}(500, 100).$$

Second, because this model estimates `sigma` explicitly, it receives its own weakly informative prior on the log scale, centered at the simulated Gaussian standard deviation of 60 ms:

$$\log(\sigma) \sim \mathcal{N}(\log(60), 0.5).$$

The coefficient for `Blur` on `mu` and the exponential-component priors are unchanged from the default model; the latter now simply attach to `tau` rather than `beta`, reflecting the difference in labels rather than any difference in the underlying quantity.

The model is then fit using:

```

364 1  fit_brms_new <- brm(
365 2    bf(
366 3      rt ~ Blur,      # mu submodel: here mu IS the Gaussian location
367 4      tau ~ Blur,     # exponential mean (the tail), allowed to vary
368 5      sigma ~ 1       # constant Gaussian SD
369 6    ),
370 7    data = d,
371 8    prior = priors_classical,
372 9    family = exg_family,      # custom classical ex-Gaussian family
373 10   stanvars = exg_stanvars,  # supplies its Stan implementation
374 11   chains = 2,
375 12   iter = 2000,
376 13   warmup = 500,
377 14   backend = "cmdstanr",

```

```

378 15   cores = 4,
379 16   seed = 1234
380 17   )

```

381 *Estimating the Gaussian-Location Parameter*

382 For the classical model, estimates from the primary regression formula correspond to the
 383 Gaussian-location parameter μ :

```

384 1 # Classical model: per-condition mu, now the Gaussian-location parameter.
385 2 mu_estimates <- estimate_means(
386 3   fit_brms_new,
387 4   by = "Blur",
388 5   predict = "mu",
389 6   centrality = "mean"
390 7   )

```

391 **Table 6**

392 *Estimated Gaussian-location parameter (μ) by condition from the classical model.*

	Blur	Mu	95% CrI	pd
393	Clear	477.0	[453.1, 502.8]	1
394	Low-Blur	545.4	[523.6, 567.7]	1

396 The estimates are approximately 500 ms for Clear and 550 ms for Low-Blur.

397 The contrast between the two can be obtained using:

```

398 1 # Classical model: Blur contrast in the Gaussian-location parameter mu.
399 2 mu_difference <- estimate_contrasts(
400 3   fit_brms_new,
401 4   contrast = "Blur",
402 5   predict = "mu",
403 6   centrality = "mean"
404 7   )

```

Table 7

Blur contrast in the Gaussian-location parameter (μ) from the classical model.

Contrast	Mean	95% CrI	pd
Low-Blur - Clear	68.4	[40.3, 96.3]	1

The posterior mean will not necessarily equal exactly 50 ms because of sampling variability. However, the 95% credible interval should contain the population value of 50 ms when the model successfully recovers the simulated effect.

Estimating the Exponential Component

Condition-specific estimates of τ can be obtained by requesting predictions for the `tau` distributional parameter:

```
1 # Classical model: per-condition tau (the exponential component).
2 tau_estimates <- estimate_means(
3   fit_brms_new,
4   by = "Blur",
5   predict = "tau",
6   centrality = "mean"
7 )
```

Table 8

Estimated exponential component (τ) by condition from the classical model.

Blur	Mean	95% CrI	pd
Clear	104.0	[77.2, 135.9]	1
Low-Blur	57.2	[36.7, 81.1]	1

The estimates should be close to 100 ms for Clear and 50 ms for Low-Blur.

The corresponding contrast can be obtained using:

```
1 # Classical model: Blur contrast in tau (the exponential component).
2 tau_difference <- estimate_contrasts(
3   fit_brms_new,
4   contrast = "Blur",
5   predict = "tau",
```

```
6      centrality = "mean"
7  )
```

Table 9

Blur contrast in the exponential component (τ) from the classical model.

Contrast	Mean	95% CrI	pd
Low-Blur - Clear	-46.8	[-79.1, -16.1]	0.999

The results seen in Table 9 are very similar to the default parameterization. This example illustrates the central difference between the two parameterizations. The default {brms} ex-Gaussian model directly estimates the expected response and labels that parameter μ . The exponential component is labeled `beta`, although it corresponds to the classical τ parameter. In contrast, the {brms.exgaussian} model uses μ for the Gaussian-location parameter and τ for the exponential mean, consistent with the classical parameterization commonly used in reaction-time research.

Real Example

To demonstrate how these parameterization differences play out in empirical research, we re-analyze Experiment 1 from Angele et al. (2023). In their study, participants completed a masked identity priming lexical decision task across two prime exposure durations (33 ms vs. 50 ms) and two priming conditions (Identical vs. Unrelated). Angele et al. (2023) set out to test whether masked priming influences early encoding processes—reflected in a shift of the Gaussian location parameter μ —without altering the exponential tail parameter τ (consistent with Forster’s Forster, 1998 savings hypothesis).

To fit their ex-Gaussian models, Angele et al. (2023) used the built-in `exgaussian()` family in {brms}. As established above, the primary formula in default {brms} models the overall distributional mean $E(RT) = \mu_{\text{Gaussian}} + \tau$, despite assigning the label μ to this parameter. In their manuscript, Angele et al. (2023) interpreted regression effects on μ as changes in the location of the Gaussian component μ .

Here, we apply the classical ex-Gaussian parameterization using {brms.exgaussian}, modeling reaction times with maximal random intercepts and random slopes for participants (`source`) and items (`Target`) across both μ (Gaussian location) and τ (exponential mean).

Prior specification for the classical model:

```

462 1  priors_classical <- c(
463 2    # Gaussian location mu: Intercept & effects
464 3    prior(normal(500, 100), class = "Intercept"),
465 4    prior(normal(0, 100), class = "b", coef = "Condition1"),
466 5    prior(normal(0, 100), class = "b", coef = "PrimeDuration1"),
467 6    prior(normal(0, 100), class = "b", coef = "Condition1:PrimeDuration1"),
468 7
469 8    # Exponential tail tau (on log scale): Intercept & effects
470 9    prior(normal(log(100), 0.5), class = "Intercept", dpar = "tau"),
471 10   prior(normal(0, 0.5), class = "b", dpar = "tau"),
472 11
473 12   # Gaussian SD sigma (on log scale): Intercept
474 13   prior(normal(log(60), 0.5), class = "Intercept", dpar = "sigma")
475 14 )

```

476 *Initial Values Specification*

477 When fitting complex hierarchical ex-Gaussian models with random effects on parameters that
478 use log link functions (such as τ and σ), Stan's default unconstrained random initialization (in $[-2, 2]$)
479 can cause severe numerical instability. On the log scale, standard deviations randomly drawn near
480 upper bounds (e.g., $e^2 \approx 7.38$) can generate extreme individual-level parameter deviations (e.g., $e^{10} \approx$
481 22,000 ms), causing MCMC chains to immediately reject initial values (log probability evaluates
482 to $\log(0)$) or fail to start.

483 To prevent initial value rejection and ensure reliable sampling across all chains, we supply an
484 explicit initialization function (`init_classical`) that sets reasonable starting values for all model
485 parameters:

```

486 1  # Subset data to determine random effect dimensions
487 2  data_fit <- expl_data_to_include %>%
488 3    filter(corr == 1 & StimulusType == "Word" & rt > 250 & rt < 1800)
489 4
490 5  N_source <- length(unique(data_fit$source))

```

```

491 6   N_target <- length(unique(data_fit$Target))
492 7
493 8   # Function generating stable initial values for all Stan parameters
494 9   init_classical <- function() {
495 10    list(
496 11      Intercept = 500,          # Gaussian location (identity link)
497 12      Intercept_sigma = log(60), # Gaussian SD (log link)
498 13      Intercept_tau = log(100), # Exponential tail (log link)
499 14      b = rep(0, 3),           # Fixed effects on mu
500 15      b_tau = rep(0, 3),       # Fixed effects on tau
501 16      sd_1 = rep(30, 4),       # Participant SDs for mu (ms)
502 17      sd_2 = rep(30, 4),       # Item SDs for mu (ms)
503 18      sd_3 = rep(0.1, 4),      # Participant SDs for tau (log scale, small to
504 prevent overflow)
505 19      sd_4 = rep(0.1, 4),      # Item SDs for tau (log scale, small to prevent
506 overflow)
507 20      z_1 = matrix(0, nrow = 4, ncol = N_source), # Participant random effects
508 for mu
509 21      z_2 = matrix(0, nrow = 4, ncol = N_target), # Item random effects for mu
510 22      z_3 = matrix(0, nrow = 4, ncol = N_source), # Participant random effects
511 for tau
512 23      z_4 = matrix(0, nrow = 4, ncol = N_target), # Item random effects for tau
513 24      L_1 = diag(4),           # Cholesky factor for participant random effects
514 (mu)
515 25      L_2 = diag(4),           # Cholesky factor for item random effects (mu)
516 26      L_3 = diag(4),           # Cholesky factor for participant random effects
517 (tau)
518 27      L_4 = diag(4)            # Cholesky factor for item random effects (tau)
519 28    )
520 29  }

```

The classical ex-Gaussian model for Experiment 1 is specified and fit as follows:

```

1  # Setup custom classical ex-Gaussian family from {brms.exgaussian}
2  exg_stanvars <- exgaussian2_stancode()
3  exg_family <- exgaussian2()
4
5  # Fit classical ex-Gaussian hierarchical model to Experiment 1 RT data
6  blmm_exp1_classical_rt <- brm(
7    data = data_fit,
8    formula = bf(
9      rt ~ Condition * PrimeDuration + (1 + Condition * PrimeDuration|source) + (1
10 + Condition * PrimeDuration|Target),
11      tau ~ Condition * PrimeDuration + (1 + Condition * PrimeDuration|source) +
12      (1 + Condition * PrimeDuration|Target),
13      sigma ~ 1
14    ),
15    warmup = 1000,
16    iter = 5000,
17    chains = 4,
18    prior = priors_classical,
19    family = exg_family,
20    stanvars = exg_stanvars,
21    inits = init_classical,
22    cores = 4,
23    backend = "cmdstanr",
24    threads = threading(2)
25  )

```

Depending on the hardware used, this model may take more than 30 minutes to fit. Because of this, it is recommended to fit it once and then save the fitted model object to a file for later use.

Gaussian-Location Parameter (μ)

Under the classical parameterization, the primary regression formula directly estimates the Gaussian location parameter μ . [Table 10](#) displays the estimated posterior means for μ across priming

conditions and exposure durations, and Table 11 presents the model's population-level predictor effects (main effect of relatedness, main effect of prime duration, and their interaction) alongside simple priming contrasts (Unrelated minus Identical) at each prime duration.

Table 10

Estimated Gaussian-location parameter (μ) by condition and prime duration for Experiment 1 from Angele et al. (2023) under the classical model.

Condition	PrimeDuration	Mean	SE	95% CrI
Identical	33 ms	509.6	7.5	[494.7, 524.1]
Unrelated	33 ms	529.6	7.4	[514.7, 544.2]
Identical	50 ms	498.0	7.6	[482.9, 512.9]
Unrelated	50 ms	535.0	7.4	[520.3, 549.4]

Table 11

Fixed effects and priming contrasts in the Gaussian-location parameter (μ) for Experiment 1 from Angele et al. (2023) under the classical model.

Effect	Estimate	SE	95% CrI
Main Effect of Relatedness	28.5	1.9	[24.7, 32.4]
Main Effect of Prime Duration	-3.1	1.7	[-6.5, 0.2]
Relatedness $\times$ Prime Duration Interaction	17.1	3.4	[10.5, 23.8]
Priming Effect at 33 ms	20.0	2.6	[14.9, 25]
Priming Effect at 50 ms	37.1	2.6	[31.9, 42.2]

As shown in Table 11, under the classical parameterization, the true Gaussian location μ shows a robust overall main effect of relatedness (28.5, 95% CrI [24.7, 32.4]). Crucially, the main effect of prime duration on μ is **non-credible** (-3.1, 95% CrI [-6.5, 0.2], with the 95% CrI containing zero), in contrast to the location shift reported under the default parameterization. The model also reveals a credible positive interaction between relatedness and prime duration (17.1, 95% CrI [10.5, 23.8]), indicating that the priming effect on μ was significantly larger at 50 ms (37.1, 95% CrI [31.9, 42.2]) than at 33 ms (20.0, 95% CrI [14.9, 25.0]).

Exponential Component (τ)

Table 12 displays the estimated exponential mean τ across conditions, and Table 13 presents the corresponding fixed effects and priming contrasts.

Table 12

Estimated exponential component () by condition and prime duration for Experiment 1 from Angele et al. (2023) under the classical model.

Condition	PrimeDuration	Mean	SE	95% CrI
Identical	33 ms	111.7	5.2	[101.7, 121.9]
Unrelated	33 ms	113.7	5.7	[102.9, 125]
Identical	50 ms	101.5	5.4	[91, 112.2]
Unrelated	50 ms	105.5	5.4	[95.3, 116.2]

Table 13

Fixed effects and priming contrasts in the exponential component () for Experiment 1 from Angele et al. (2023) under the classical model.

Effect	Estimate	SE	95% CrI
Main Effect of Relatedness	3.0	2.5	[-1.9, 8.1]
Main Effect of Prime Duration	-9.2	2.9	[-14.8, -3.5]
Relatedness $\times$ Prime Duration Interaction	2.1	5.4	[-8.5, 12.6]
Priming Effect at 33 ms	2.0	3.8	[-5.4, 9.5]
Priming Effect at 50 ms	4.1	3.6	[-2.9, 11.2]

For the exponential component τ , the main effect of relatedness is small and non-credible (3.0, 95% CrI [-1.9, 8.1]), as is the interaction term (2.1, 95% CrI [-8.0, 12.8]; log scale $b = 0.022$). Consequently, simple priming differences at 33 ms (2.0, 95% CrI [-5.4, 9.5]) and 50 ms (4.1, 95% CrI [-2.9, 11.2]) remain non-credible, consistent with Forster's (1998) savings hypothesis. However, the model reveals a credible main effect of prime duration (-9.2, 95% CrI [-14.7, -3.5]; log scale $b = -0.085$, 95% CrI [-0.137, -0.032]), showing that increasing prime duration from 33 ms to 50 ms moderately compressed the exponential tail parameter τ (from ~ 113 ms to ~ 104 ms).

Comparison with Angele et al. (2023)

Comparing these classical ex-Gaussian estimates with those originally reported by Angele et al. (2023) highlights the practical consequences of model parameterization for theoretical inferences (Table 14).

Table 14

Side-by-side comparison of estimated effects from Experiment 1 under the default brms parameterization ($E(RT)$) versus the classical parameterization (Gaussian location and exponential tail).

Effect	Default brms $E(RT)$	Classical μ	Classical τ
Main Effect of Relatedness	31.5 [27.7, 35.3]	28.5 [24.7, 32.4]	3.0 [-1.9, 8.1]
Main Effect of Prime Duration	-10.9 [-14.5, -7.3]	-3.1 [-6.5, 0.2]	-9.2 [-14.7, -3.5]
Relatedness $\times$ Prime Duration Interaction	19.3 [12.6, 26.0]	17.1 [10.5, 23.8]	2.1 [-8.0, 12.8]
Priming Effect at 33 ms	22.0	20.0 [14.9, 25.0]	2.0 [-5.4, 9.5]
Priming Effect at 50 ms	41.0	37.1 [31.9, 42.2]	4.1 [-2.9, 11.2]

- Parameter Interpretation:** Angele et al. (2023) reported fixed effects for relatedness, prime duration, and their interaction from default {brms} under the assumption that `mu` represented the Gaussian location component μ_{Gaussian} . In reality, `brms` default `mu` estimates $E(RT) = \mu_{\text{Gaussian}} + \tau$. Consequently, Angele et al. (2023) reported overall distributional mean differences under the label of Gaussian location shifts.
- Divergent Conclusions on Prime Duration:** A critical theoretical discrepancy emerges regarding the main effect of prime duration. Based on their default {brms} model, Angele et al. (2023) concluded that increasing prime exposure duration from 33 ms to 50 ms produced a credible downward shift in the Gaussian location parameter μ ($b = -10.9$ ms, 95% CrI [-14.5, -7.3]). However, the classical parameterization reveals that the true main effect of prime duration on μ_{Gaussian} is **non-credible** (-3.1 ms, 95% CrI [-6.5, 0.2], with the 95% CrI containing zero). Instead, the overall reduction in RTs at 50 ms prime duration was driven almost entirely by a compression of the exponential tail τ (-9.2 ms, 95% CrI [-14.7, -3.5]; log scale $b = -0.085$, 95% CrI [-0.137, -0.032]). Because default {brms} models $E(RT) = \mu_{\text{Gaussian}} + \tau$, the -10.9 ms effect reported by Angele et al. (2023) as a location shift was actually the composite sum of a non-credible μ shift (-3.1 ms) and a credible τ tail reduction (-9.2 ms).
- Priming Effects and Interaction:** For the priming manipulation, the main effect of relatedness on μ_{Gaussian} (28.5 ms, 95% CrI [24.7, 32.4]) and the positive interaction (17.1 ms, 95% CrI [10.5, 23.8]) closely mirror the $E(RT)$ effects reported by Angele et al. (2023) (main effect ~ 31.5 ms, interaction ~ 19.3 ms). The estimates for the two parameterizations align closely for the priming effect because τ showed no credible priming differences (3.0 ms, 95% CrI [-1.9, 8.1]). When $\Delta\tau \approx$

0, changes in the distributional mean $\Delta E(RT)$ track changes in the Gaussian location $\Delta\mu$ nearly 1-to-1.

4. **Methodological Takeaway:** This empirical comparison demonstrates that misinterpreting default {brms} parameters can lead to subtle misattributions regarding *which* distributional component is altered by an experimental manipulation. In Angele et al. (2023), a tail-shortening effect of prime duration ($\Delta\tau = -9.2$ ms) was attributed to the Gaussian location parameter (μ). Adopting the classical parameterization resolves this misattribution and ensures that theoretical claims align with the underlying distributional parameters.

Conclusion

With this commentary, I hope to encourage researchers to use the appropriate ex-Gaussian parameterization when modeling reaction-time data with the {brms} package. The ex-Gaussian distribution provides a powerful framework for moving beyond analyses of the mean by separating differences in the central portion of the reaction-time distribution from differences in its right tail.

However, this flexibility also requires careful attention to model parameterization. In particular, researchers should not assume that parameters with familiar labels necessarily have the same interpretation across software implementations. Researchers should therefore examine the statistical definition and underlying implementation of a model before fitting it and interpreting its parameters. This is especially important when using flexible modeling software that may adopt parameterizations that differ from those commonly used in a particular research literature. I hope this commentary serves both as a practical guide to fitting the classical ex-Gaussian model in {brms} and as a broader reminder that convenient software defaults should not replace careful consideration of what a model actually estimates.

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